



**Pharm***Achieve*

# HOSPITAL PHARMACY PRACTICE

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

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**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

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**ADU** Automated Dispensing Unit

**BPMH** Best Possible Medication History

**BSA** Body Surface Area

**C-MAR** Computerized Medication Administration Record

**CPOE** Computerized Physician/Provider Order Entry

**DTP** Drug Therapy Problem

**DUE** Drug Use Evaluation

**DVT** Deep vein thrombosis

**ED** Emergency Department

**EMR** Electronic Medical Record

**HCP** Healthcare Professional

**IESA** Indication, Efficacy, Safety, Adherence

**ISMP** Institute for Safe Medication Practices

**IM** Intramuscular

**IV** Intravenous

**MAC** Medication Advisory Committee

**MAR** Medication Administration Record

**Med Rec** Medication Reconciliation

**NaCl** Sodium Chloride

**NAPRA** National Association of Pharmacy Regulatory Authorities

**PO** Oral

**PPE** Personal Protective Equipment

**P&T** Pharmacy and Therapeutics

**Q** Quaque (Every)

**Rx** Prescription

**SC** Subcutaneous

**STAT** Statim (Immediately)

**TDM** Therapeutic Drug Monitoring

**Vs.** Versus

**WBC** White Blood Cells

**WHO** World Health Organization

# UNDERSTANDING HOSPITAL PRACTICE

“Similar to those performed by community pharmacists, hospital pharmacists interact closer with prescribers and other allied health to promote the rational prescribing of drugs, have greater access to medical records (i.e. laboratory results) thus being able to better influence drug selection and dosing, monitor therapeutic response, drug compliance, and adverse drug reactions.

Hospital pharmacists also participate in policy-making committees (i.e. Pharmacy and Therapeutics Committee), educate other healthcare professionals about appropriate drug use, partake in implementation of clinical trials, and control hospital manufacturing and procurement of pharmaceutical products.”

Source: *World Health Organization (WHO)*



# INPATIENT vs. OUTPATIENT

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## OUTPATIENT PHARMACIST

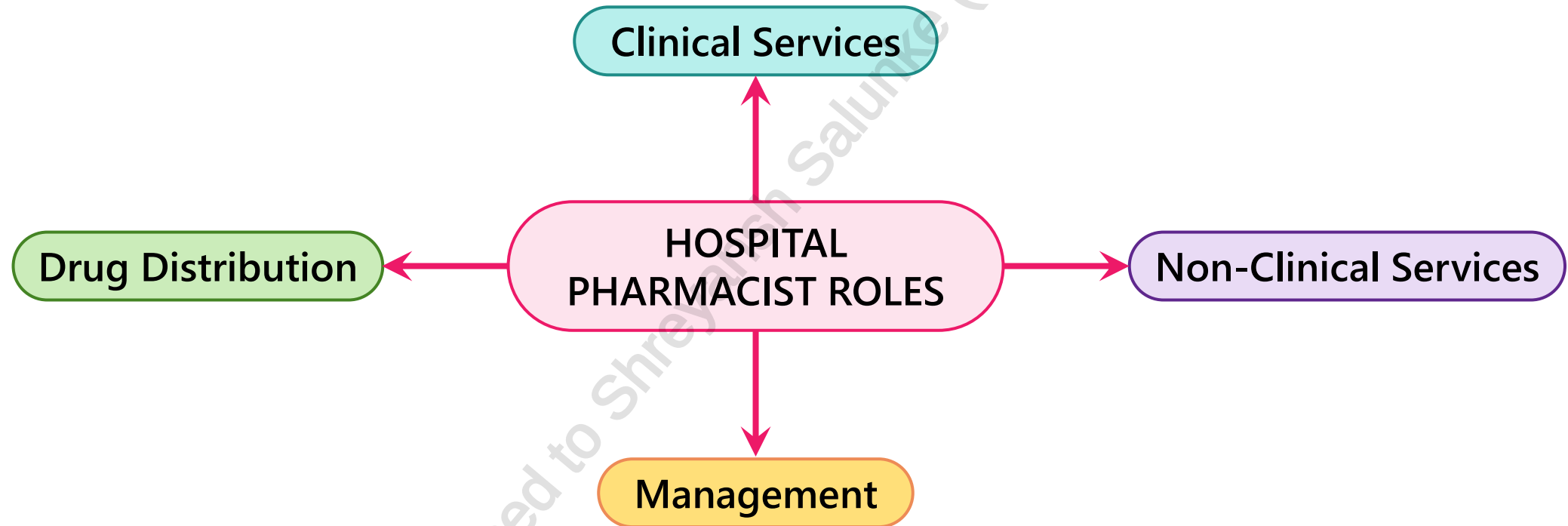
A pharmacist who works in a community pharmacy located within a hospital OR in a hospital clinic (i.e. oncology, dialysis) that serves patients not admitted to the hospital

## INPATIENT PHARMACIST

A pharmacist who works as a part of the hospital's allied health team to assist in providing pharmaceutical care to patients admitted to the hospital

# INPATIENT HOSPITAL PHARMACIST ROLES

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# HOSPITAL PHARMACY PRACTICE MODELS

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## Drug distribution-centred model

Pharmacists' function is drug distribution with limited clinical services (i.e. interventions occur secondary to order verification)

## Separate clinical and distributive practice model

2 distinct groups of pharmacists, distributive and clinical, that solely perform their designated roles

## Clinical practice-centred model

Perform clinical duties with less than 20% in distribution. Pharmacy technicians and automation manage drug distribution

## Integrated drug distribution/clinical practice model

50% distribution and 50% clinical duties by all pharmacists

# CLINICAL SERVICES

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- Checking medication orders
- Completing medication reconciliation
- Developing pharmaceutical care plans
- Practicing antimicrobial stewardship
- Therapeutic drug monitoring (TDM)
- Attending patient care rounds
- Providing patient education
- Calculating appropriate medication doses, infusion rates and IV to PO conversions
- Answering drug information questions



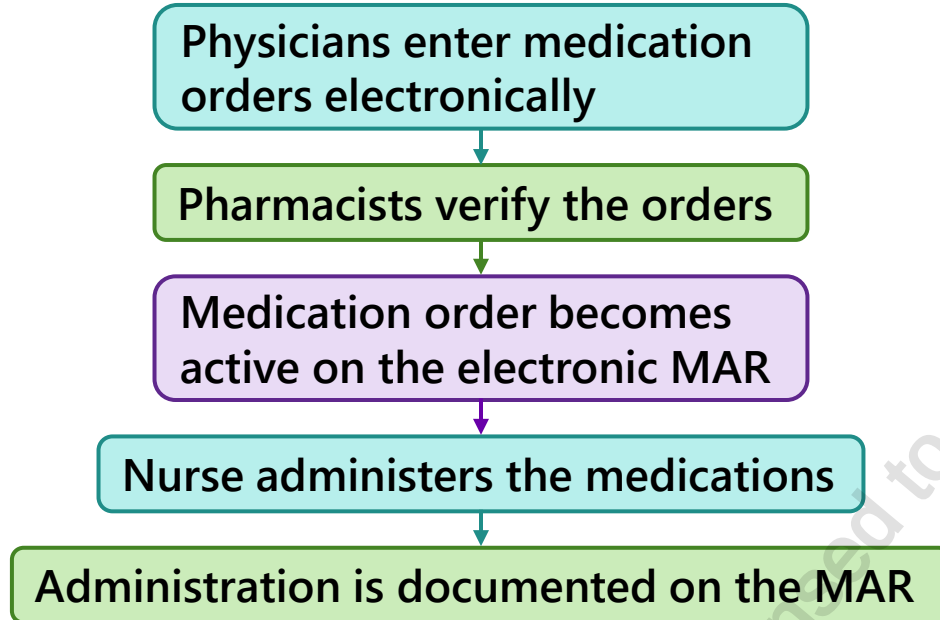
Note that this list is not comprehensive and can vary from institution to institution



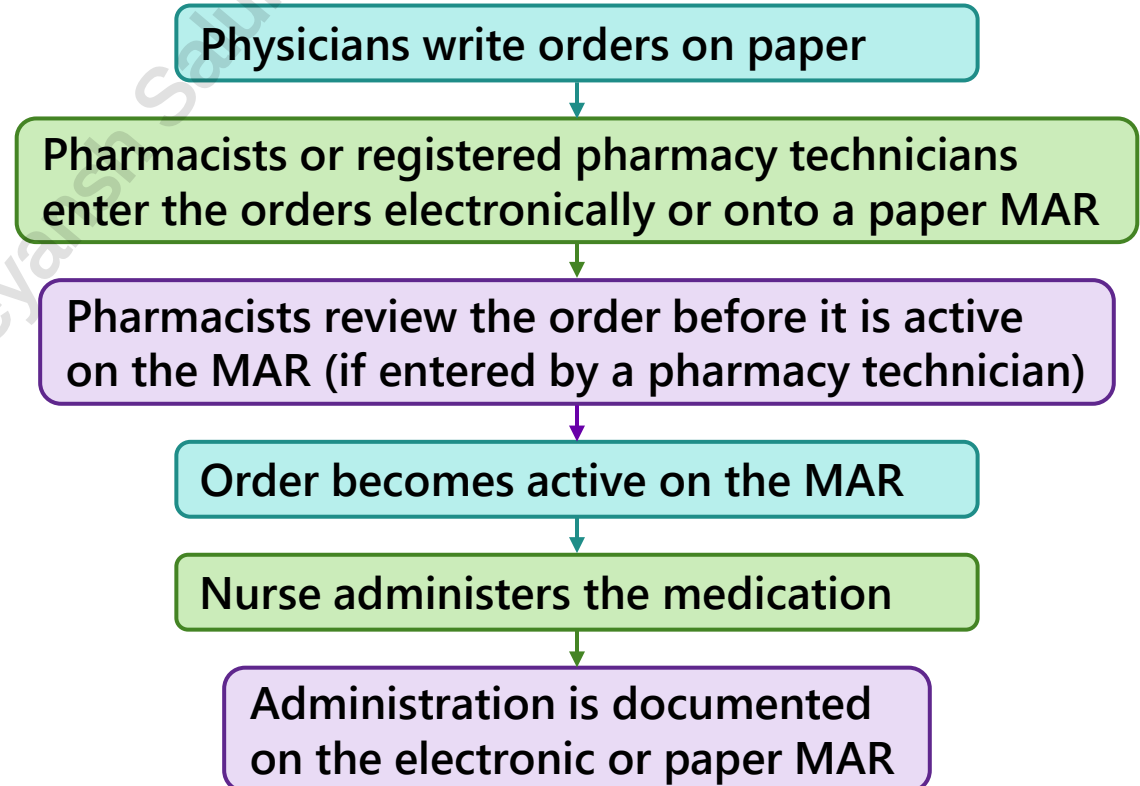
# CHECKING MEDICATION ORDERS - WORKFLOW

Hospital pharmacists must review all medication orders for indication, efficacy, safety and adherence (IESA) before administration

## Computerized provider/physician order entry (CPOE) institutions



## Paper-based institutions



# CHECKING MEDICATION ORDERS CLINICAL ASSESSMENT

## INDICATION

- Is each medication indicated?
- Is there a duplicate medication?
- Is drug therapy indicated for an untreated medical condition?

## SAFETY

- Are there drug interactions?
- Are plasma levels within target range (if applicable)?
- Is the dose too high?
- Is the patient's hepatic and renal function sufficient to clear the medication?
- Is the patient at high risk of adverse effects?
- Is the patient allergic to the medication?

## EFFICACY

- Is the medication effective?
- Is the dose effective?
- Are plasma levels within target range (if applicable)?
- Are there any drug interactions?
- Are there laboratory tests to indicate efficacy? (e.g. microorganism growth eliminated from blood cultures)

## ADHERENCE

- Is there sufficient IV access? Is a central line present if needed?
- Can IV medications be switched to oral medications?
- Can the medication be crushed if administered via a tube?

# ORDER VERIFICATION

| Inpatient Orders                   |                                                                                                           |      |                               |          |            |         |                                    |          |                     |  |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|------|-------------------------------|----------|------------|---------|------------------------------------|----------|---------------------|--|
| Name                               | Order Description                                                                                         | O... | Facility                      | FIN      | Queue Time | Status  | Ordering Physician                 | Age      | Last Unverified Act |  |
| PaedsOnc, 1732 Ind Patient         | piperacillin-tazobactam (piperacillin-tazobactam (paeds)) 7,200 mg, syringe, IV, q6 hours, for: 3...      |      | LHSC-VC                       | 40106712 | 971 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 16 years | Order               |  |
| PaedsOnc, 1732 Ind Patient         | tobramycin 180 mg = 4.5 mL, injection, IV, q8 hours, infuse over 60 min, for: 48 hr, Start: 2023/01/25... |      | LHSC-VC                       | 40106712 | 971 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 16 years | Order               |  |
| PaedsOnc, 1732 Ind Instructor      | piperacillin-tazobactam (piperacillin-tazobactam (paeds)) 7,200 mg, syringe, IV, q6 hours, for: 3...      |      | LHSC-VC                       | 40106713 | 971 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 16 years | Order               |  |
| PaedsOnc, 1732 Ind Instructor      | tobramycin 180 mg = 4.5 mL, injection, IV, q8 hours, infuse over 60 min, for: 48 hr, Start: 2023/01/25... |      | LHSC-VC                       | 40106713 | 971 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 16 years | Order               |  |
| Obstetrics Post Partum, Vaginal 01 | oxytocin 20 units IV continuous, 167 mL/hr, for: 6 hr, Total volume (mL): 1,000, Start: 2023/01/26 10:... |      | LHSC-VC                       | 40106774 | 970 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 35 years | Order               |  |
| Obstetrics, Newborn 01             | phytonadione 1 mg, injection, IM, ONCE, Start: 2023/02/10 8:42:00 EST, Stop: 2023/02/10 8:42:00 ES...     |      | LHSC-VC                       | 40106770 | 956 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 2 years  | Order               |  |
| Obstetrics, Newborn 02             | phytonadione 1 mg, injection, IM, ONCE, Start: 2023/02/10 8:52:00 EST, Stop: 2023/02/10 8:52:00 ES...     |      | LHSC-VC                       | 40106771 | 956 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 2 years  | Order               |  |
| Obstetrics, Newborn 03             | phytonadione 1 mg, injection, IM, ONCE, Start: 2023/02/10 8:54:00 EST, Stop: 2023/02/10 8:54:00 ES...     |      | LHSC-VC                       | 40106772 | 956 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 2 years  | Order               |  |
| Obstetrics, Newborn 04             | phytonadione 1 mg, injection, IM, ONCE, Start: 2023/02/10 8:57:00 EST, Stop: 2023/02/10 8:57:00 ES...     |      | LHSC-VC                       | 40106773 | 956 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 2 years  | Order               |  |
| Paeds, Instructor                  | dextrose 5%-sodium chloride 0.45%-potassium chloride 20 mmol/L 1,000 mL IV continuous, 2...               |      | LHSC-VC                       | 40106728 | 945 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 5 years  | Order               |  |
| Garcia, WD ED 01                   | Pharmacy Med Order Review 1 ea, -, ORAL, as directed, for: 24 hr, Stop: 2023/01/17 9:10:00 EST            |      | Wingham and District Hospital | 50100496 | 981 days   | Ordered | System, System                     | 76 years | Order               |  |
| Johnson, UH ED 01                  | ibuprofen 600 mg, tab, ORAL, ONCE, Start: 2023/01/18 8:30:00 EST, Stop: 2023/01/18 8:30:00 EST            |      | LHSC-UC                       | 50100513 | 979 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 28 years | Modify              |  |
| Johnson, UH ED 01                  | Pharmacy Med Order Review 1 ea, -, ORAL, as directed, for: 24 hr, Stop: 2023/01/19 8:33:00 EST            |      | LHSC-UC                       | 50100513 | 979 days   | Ordered | System, System                     | 28 years | Order               |  |
| Adams, UH ED 02                    | ibuprofen 600 mg, tab, ORAL, ONCE, Start: 2023/01/18 8:45:00 EST, Stop: 2023/01/18 8:45:00 EST            |      | LHSC-UC                       | 50100514 | 979 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 28 years | Modify              |  |
| Adams, UH ED 02                    | Pharmacy Med Order Review 1 ea, -, ORAL, as directed, for: 24 hr, Stop: 2023/01/19 8:37:00 EST            |      | LHSC-UC                       | 50100514 | 979 days   | Ordered | System, System                     | 28 years | Order               |  |
| Johnson, VH AED 01                 | Pharmacy Med Order Review 1 ea, -, ORAL, as directed, for: 24 hr, Stop: 2023/01/19 8:44:00 EST            |      | LHSC-VC                       | 50105515 | 979 days   | Ordered | System, System                     | 28 years | Order               |  |
| Johnson, VH AED 01                 | ibuprofen 600 mg, tab, ORAL, ONCE, Start: 2023/01/18 8:45:00 EST, Stop: 2023/01/18 8:45:00 EST            |      | LHSC-VC                       | 50105515 | 979 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 28 years | Modify              |  |
| Adams, VH AED 02                   | Pharmacy Med Order Review 1 ea, -, ORAL, as directed, for: 24 hr, Stop: 2023/01/19 8:50:00 EST            |      | LHSC-VC                       | 50105516 | 979 days   | Ordered | System, System                     | 28 years | Order               |  |
| Adams, VH AED 02                   | ibuprofen 600 mg, tab, ORAL, ONCE, Start: 2023/01/18 9:00:00 EST, Stop: 2023/01/18 9:00:00 EST            |      | LHSC-VC                       | 50105516 | 979 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 28 years | Modify              |  |
| Johnson, SE ED 01                  | ibuprofen 600 mg, tab, ORAL, ONCE, Start: 2023/01/18 9:00:00 EST, Stop: 2023/01/18 9:00:00 EST            |      | St. Thomas Elgi               | 50100518 | 979 days   | Ordered | Test, Doc (Physician -DONT CHANGE) | 28 years | Modify              |  |
| Johnson, SE ED 01                  | Pharmacy Med Order Review 1 ea, -, ORAL, as directed, for: 24 hr, Stop: 2023/01/19 9:03:00 EST            |      | St. Thomas Elgi               | 50100518 | 979 days   | Ordered | System, System                     | 28 years | Order               |  |

# ORDER VERIFICATION

PharmNet, Patient 01

Enter a patient name:

Demographics

Height

170 cm

Result Dt/Tm

2025/09/22 07...

Weight

75.1 kg

Result Dt/Tm

2025/09/22 07...

Creatinine

95

Result Dt/Tm

2025/09/22 07...

BSA

1.88 m<sup>2</sup>

BSA Method

Mosteller

IBW

61.44 kg

IBW Method

Devine

CrCl (est.)

MISSING DATA

CrCl Method

Cockcroft-Gault ...

Patient Note[]

Click here to view/edit the patient...

Reason For Visit [1]

Shortness of Breath

Allergies [4]

Click here to add a new allergy...

Cherry

Cat dander

ginger

sulfa drugs

Medications

amLODIPine 5 mg / 1 tab ORAL ...

dalteparin 5,000 units / 0.2 mL S...

diltiazem ER 120 mg / 1 cap O...

fentaNYL (25 mcg/hr) 1 patch tra...

furosemide 40 mg / 1 tab ORAL ...

levodopa 100 mg-carbidopa 25 ...

levodopa-carbidopa 200 mg-50 ...

metoprolol 200 mg ORAL daily

pregabalin 75 mg / 1 cap ORAL ...

sennosides 8.6 mg / 1 tab ORAL ...

venlafaxine ER 225 mg / 3 cap O...

zopiclone 5 mg / 1 tab ORAL bed...

Rx Clinical Note[]

MRN:   
FIN: 4

Age: 69 Years

Sex: Female

Location: U-4MED3 / A4-230 A

Admitted: 2022/12/19 07:00 EST

Physician: Test, Doc (Physician -DONT CHANGE)

Rx Clinical Note:

Verify Med Order

Drug:  

| Drug                | Dose         | Ordered As |
|---------------------|--------------|------------|
| amLODIPine          | 5 mg / 1 tab | amLODIPine |
| amLODIPine 5 mg TAB |              |            |

Route: ORAL

Frequency: daily

PRN doses: ☐

PRN reason: (None)

Physician: Test, Doc (Physician -DONT CHA

Duration: (None)

Start date: Original 2023/01/24 08:00 EST

Stop date: \*\*\*\*/\*\*/\*\*

Time: 08:00

Stop type: No

Previous scheduled administration: E... 2025/09/23 08:00

Next administration: EST 2023/01/25 08:00

Following: EST 2023/01/26 08:00

Remaining doses:

Order comments:

Product notes:

Dosage form: tab

Communication type: Signed Paper Order

Order priority: Routine

Product...

Dispense category: ROBOTL

Dispense from location: U Main Pharmacy

Initial doses: 0

Initial quantity: 0

Printing...

Billing formula: LHSC

Price: \$2.00

Cost: \$2.00

Comments...

Alert History...

Rx Intervention

☐ Patient's own med

☒ Auto calculate initial dose

Reject

OK

Cancel

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12

[illegible]



# MEDICATION RECONCILIATION

A Best Possible Medication History (BPMH) is a complete list of medications a patient is taking, including:

- Prescription medications (oral, inhalers, eye drops, patches, creams, etc.)
- Non-prescription medications (over-the-counter products, vitamins, herbal products, natural health products)
- Sample medications
- Recreational drug use



The medication name, dose, route and frequency must be documented. It is imperative to use at least two sources of information to complete a BPMH

Medication reconciliation occurs during transitions of care (i.e. changing setting, practitioner, or level of care). Medication reconciliation includes:

- Comparing medication orders to the BPMH
- Identifying discrepancies
- Making clinical decisions based on the comparison of the two lists
- Discrepancies and/or therapeutic concerns need to be communicated to an appropriate healthcare provider so they can be addressed

# MEDICATION RECONCILIATION

## ADMISSION MED REC

- Upon admission to the hospital, new medications may be ordered (medications the patient does not take at home). These additional medications may be intended to treat the patient's reason for admission or may be "standard" hospital orders. These include:
  - Deep vein thrombosis (DVT) prophylaxis (e.g. dalteparin or enoxaparin)
  - As needed pain, antiemetic and laxative medications

## TRANSFER MED REC

- Transfer can occur within an institution if a patient moves between levels of care or can occur when a patient is moved from one institution to another

## DISCHARGE MED REC

- DVT prophylaxis and PRN hospital orders are typically stopped upon discharge, and the new home medication list should be reviewed with the patient and sent to their community pharmacy



# DEVELOP PHARMACEUTICAL CARE PLANS

## Pharmaceutical Care Plan

A treatment plan that includes goals of therapy, interventions to prevent or resolve DTPs and follow-up/monitoring

IESA review of each medication order to identify DTPs

Care plan is developed including goals of therapy, interventions and monitoring

Care plan is discussed with the appropriate healthcare professional

DTPs are resolved and/or monitoring plan put in place

Documentation

# DEVELOPING PHARMACEUTICAL CARE PLANS

- Ensure the goals of therapy are feasible within the patient's anticipated duration of stay
- Prioritize 1-2 main drug therapy problems (e.g. medications contributing to the reason for admission)
- Care plans may change based on the clinical status and needs of the patient

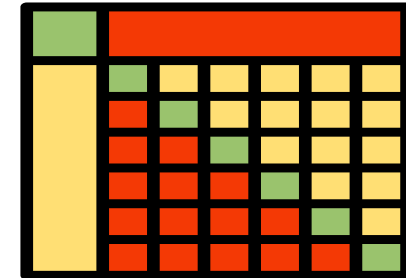
Examples:

| Background                                                                                                                                                                                                                    | DTP                                       | Care Plan                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient is admitted after a motor vehicle accident. Bloodwork was done and potassium level was found to be 5.2 (reference range 3.5-5.0 mmol/L)<br>Home medications include vitamin D, spironolactone, metformin and ramipril | Patient has high potassium level          | <ul style="list-style-type: none"><li>• Hold patient's home ramipril and spironolactone</li><li>• Continue to monitor potassium level daily</li><li>• Monitor BP daily</li><li>• Restart when appropriate</li></ul> |
| An 88-year-old patient is admitted following a fall.<br>Home medications include atorvastatin, quetiapine, tamsulosin, lorazepam, and dimenhydrinate.                                                                         | Polypharmacy<br>→ increased risk of falls | <ul style="list-style-type: none"><li>• Check orthostatic vitals since tamsulosin can cause orthostatic hypotension</li><li>• Reassess indication for quetiapine + lorazepam + dimenhydrinate</li></ul>             |

# ANTIMICROBIAL STEWARDSHIP

- Antimicrobial Stewardship is used to improve antibiotic use and decrease the risk of antimicrobial resistance
- Antimicrobial Stewardship Programs aim to have prescribers:
  - Choose antibiotics with a spectrum of action most targeted against the known or likely microorganisms
  - Minimize the use of broad-spectrum antibiotics
- Antibiograms help guide empiric antibiotic therapy
- Cultures (e.g. blood, urine, sputum, wound, tracheal, etc.) and their sensitivities can help guide targeted antibiotic therapy

Know your antibiogram  
before recommending  
antimicrobials



Nova Scotia Health | Antimicrobial Stewardship (AMS)

# ANTIMICROBIAL STEWARDSHIP

- Culture results will state the type of culture (blood, sputum, wound, urine, etc.)
- Repeat cultures may be taken to ensure organism growth is not a contaminant
- Avoid using intermediate sensitivity antibiotics as the risk of treatment failure is high
- Sensitivities may be listed to only one medication in a drug class. The sensitivity can often be extrapolated to other medications in the same class. For example:
  - Penicillin sensitivity can be extrapolated to amoxicillin sensitivity
  - Ciprofloxacin sensitivity can be extrapolated to levofloxacin sensitivity

| Culture                            | Time of Culture | Organism                | Antibiotic    | Sensitivity |
|------------------------------------|-----------------|-------------------------|---------------|-------------|
| Right arm peripheral blood culture | Apr 15 at 14:35 | <i>Escherichia coli</i> | Penicillin    | I           |
|                                    |                 |                         | Ceftriaxone   | S           |
|                                    |                 |                         | Ciprofloxacin | R           |
|                                    |                 |                         | Gentamicin    | S           |
|                                    |                 |                         | Meropenem     | S           |

S=sensitive, I=intermediate, R=resistant

# OTHER CLINICAL RESPONSIBILITIES

## THERAPEUTIC DRUG MONITORING (TDM)

- Monitor the efficacy and safety of medications (i.e. INR, vancomycin levels, phenytoin levels, digoxin levels, etc.)
- Monitor bloodwork parameters in response to medications (i.e. WBC and neutrophils with clozapine or WBC with antibiotics)

## ATTEND PATIENT CARE ROUNDS

- Review cases with other HCPs
- Discuss DTPs and care plans

## PROVIDE PATIENT EDUCATION

- Review medications, answer questions and address concerns

## CALCULATIONS

- Dosing calculations based on weight/BSA, or TDM, and IV-to-PO dose conversions

## ANSWER DRUG INFORMATION QUESTIONS

- Questions may come from any HCP (e.g. nurse, physician, dietician, social worker, etc.), manager, or a patient

# NON-CLINICAL SERVICES

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**1**

## **Drug safety**

- Adverse drug event reporting
- Medication error audits
- Appropriate handling of hazardous drugs

**2**

## **Drug use evaluation (DUE)**

- Pharmacy & Therapeutics (P & T) Committee
- Medication Advisory Committee
- Hospital drug formulary creation
- Pre-printed order set development

**3**

## **Resource Maintenance**

- Training employees
- Medication manuals



# DRUG SAFETY

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## ADVERSE EVENT REPORTING

- Reporting any adverse drug events to Health Canada and the hospital reporting system

## MEDICATION ERROR AUDITS

- Errors are reported through the hospital's reporting system
- Errors are audited and discussed to prevent future errors

## APPROPRIATE DRUG HANDLING

- Policies and procedures for handling, preparation, dispensing, transportation, storage and disposal of cytotoxic and non-cytotoxic hazardous medications

# DRUG USE EVALUATION

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## PHARMACY AND THERAPEUTICS COMMITTEE (P&T)

- Committee that discusses medication management and hospital formulary
- Consists of physicians, nurses, pharmacists, administrators and safety officers

## HOSPITAL MEDICATION FORMULARY

- Medications approved by the P&T committee to be dispensed within the hospital
- Medications are reviewed for efficacy, safety, and cost before being added or removed from the formulary

## MEDICATION ADVISORY COMMITTEE (MAC)

- Committee that discusses medication policies and management
- Consists of pharmacy managers, clinical leaders and safety officers

## PREPRINTED ORDER SET DEVELOPMENT

- Standardized document used for quick, appropriate and safe ordering of medications, labs, imaging etc. for a specific disease state
- Created via collaboration between physicians, pharmacists, nurses, clinical informatics and quality and safety officers

# PREPRINTED ORDER SETS

01

Age: 28 years

Weight: 76 kg

Ht/Length: 180 cm

BSA:

PIN:

Visit #: 050105313

Sex: Male

Inpatient [2022/12/19 07:00 - <No - Discharge date>]

Location: V-AEM; Blue; 1

Provider: Test, Doc (Physician -DONT CHANGE)

! Behaviour Safety Alert: High

Allergies: garlic, No Known Environmental Allergies, No Known Food Allergies

Menu

Summaries ViewPoint

Oncology

Results Review

Orders + Add

Medication List + Add

Appointments

View / I&O

Form Browser

Clinical Documents/Reports

MAR

MAR Summary

ClinicalConnect

Pharmacy View

Pharmacy Referrals

Pharmacy Interventions Completed

Patient Information

Chemotherapy Summary

Blood Product Information

Clinical Notes Viewer

Allergies + Add

Quick Orders

Documents

Histories

Problems and Diagnoses

Behaviour Safety Alert Summary

LOA

Infection Control

Client Info

Orders

Medication List

Document In Plan

+ Add

Document Medication by Hx

Reconciliation \*

Check Interactions

Reconciliation Status

✓ Meds History

Admission

Disc

Orders

Medication List

Document In Plan

+ Add to Phase

Check Alerts

Comments

Start: Now

Duration: None

View All

|                                                                                                                                                                                                                                                                                                                                                                                                                            | Component                                                                   | Status | Dose ...                                                                       | Details                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------|----------------------------|
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | COMMON - Nicotine Replacement Therapy/Smoking ...                           |        |                                                                                |                            |
| Antibacterials                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                             |        |                                                                                |                            |
| If patient has received antibiotics in the last 3 months, consider alternative class of antibiotic.                                                                                                                                                                                                                                                                                                                        |                                                                             |        |                                                                                |                            |
| Simple Exacerbation: COPD without risk factors, increased cough and sputum purulence and dyspnea, Probable Pathogens: Haemophilus influenzae, Haemophilus species, Moraxella catarrhalis, Streptococcus pneumoniae. *QBP*                                                                                                                                                                                                  |                                                                             |        |                                                                                |                            |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | amoxicillin                                                                 |        | 500 mg, cap, ORAL, q8 hours, for: 5 day                                        | *QBP*                      |
| Azithromycin is a less effective agent against H. influenzae and S. pneumoniae and should be reserved for when allergy restricts use of other agents                                                                                                                                                                                                                                                                       |                                                                             |        |                                                                                |                            |
| If patient on prophylactic azithromycin, assume the bacteria will be resistant to all macrolides. *QBP*                                                                                                                                                                                                                                                                                                                    |                                                                             |        |                                                                                |                            |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | azithromycin                                                                |        | 500 mg, tab, ORAL, ONCE, for: 1 dose, Start: T;N                               | loading dose *QBP*         |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | azithromycin                                                                |        | 250 mg, tab, ORAL, daily, for: 4 dose, Start: T+1;0800                         | maintenance dose *QBP*     |
| Alternative antibiotics                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                             |        |                                                                                |                            |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | cefuroxime                                                                  |        | 500 mg, tab, ORAL, BID with food, for: 5 day                                   | *QBP*                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | doxycycline                                                                 |        | 100 mg, cap, ORAL, q12 hours, for: 5 day                                       | *QBP*                      |
| Complex Exacerbation: COPD with risk factors, as in simple plus at least one of: FEV1 less than 50 % predicted, 4 or more exacerbations per year, ischemic heart disease, use of home oxygen, chronic oral steroid use, or antibiotic use in past 3 months. Probable Pathogens: as in simple plus: klebsiella species, other gram-negatives, increased probability of beta-lactam resistance or pseudomonas species. *QBP* |                                                                             |        |                                                                                |                            |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | amoxicillin-clavulanate (amoxicillin-clavulanate 500 mg-125 mg oral tablet) |        | 1 tab, tab, ORAL, q8 hours, for: 5 day                                         | *QBP*                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | amoxicillin-clavulanate (amoxicillin-clavulanate 875 mg-125 mg oral tablet) |        | 1 tab, tab, ORAL, q12 hours, for: 5 day                                        | *QBP*                      |
| If Oral route is inappropriate/contraindicated *QBP*                                                                                                                                                                                                                                                                                                                                                                       |                                                                             |        |                                                                                |                            |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | ceftriaxone                                                                 |        | 1 g, injection, IV, q24 hours, for: 5 dose                                     | Reassess in 48 hours *QBP* |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | azithromycin                                                                |        | 500 mg, injection, IV, ONCE, infuse over 60 min                                | *QBP*                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                   | azithromycin                                                                |        | 250 mg, injection, IV, daily, infuse over 60 min, for: 4 dose, Start: T+1;0800 | *QBP*                      |
| Bronchodilators/Anticholinergic                                                                                                                                                                                                                                                                                                                                                                                            |                                                                             |        |                                                                                |                            |

# RESOURCE MAINTENANCE

## TRAINING EMPLOYEES

- Corporate orientation
- Learning modules
- Medication distribution system shadowing
- Pharmacist shadowing

## MEDICATION RESOURCES

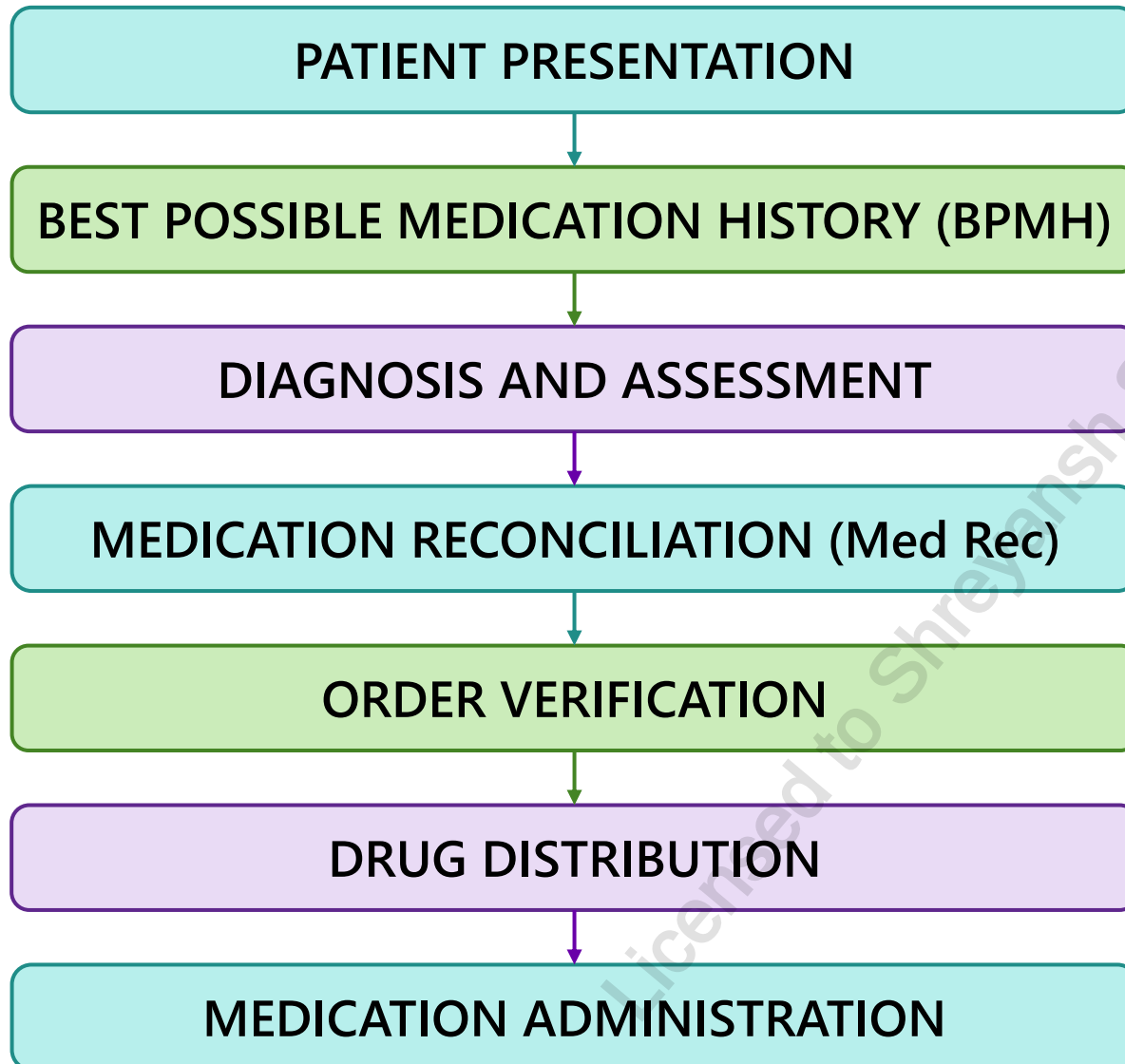
- IV resource manuals
- Pre-printed order sets
- Hospital formulary
- Drug information subscriptions

## Emergency Codes

|                                                                                       |       |                                    |                                                                                       |        |                                   |
|---------------------------------------------------------------------------------------|-------|------------------------------------|---------------------------------------------------------------------------------------|--------|-----------------------------------|
|    | Aqua  | Flood                              |    | Orange | External Disaster                 |
|    | Black | Bomb Threat / Suspicious Object    |    | Purple | Hostage Taking                    |
|    | Blue  | Cardiac Arrest / Medical Emergency |    | Red    | Fire/Smoke                        |
|    | Brown | In-facility Hazardous Spill        |    | Silver | Active Attacker<br>Active Shooter |
|   | Green | Evacuation                         |   | White  | Violent Situation                 |
|  | Grey  | Infrastructure Loss or Failure     |  | Yellow | Missing Resident                  |

# HOSPITAL DISTRIBUTION WORKFLOW

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# DISPENSING TERMINOLOGY

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## WARD STOCK

Medication supply kept on a unit or ward

## PATIENT SPECIFIC MEDICATION

Medications not stocked on the unit or ward. These medications are filled in the pharmacy and brought to the ward

## CENTRALIZED PHARMACY

One large pharmacy in the hospital which dispenses to all units or wards

## DECENTRALIZED PHARMACY

Smaller pharmacies or "satellite pharmacies" located throughout the hospital to serve certain specialized units

# MEDICATION DISPENSING

---

## Medication Dispensing Methods

- Traditional
- Automated Dispensing Units (ADUs)
- Patient Specific Unit Dose

## Compounding

- IV, IM, SC, intrathecal, oral, and epidural routes
- Sterile and non-sterile
- Hazardous and non-hazardous
- Can be dispensed traditionally or via unit doses



# MEDICATION DISPENSING METHODS

|                                      | TRADITIONAL                                                                                                                   | AUTOMATED DISPENSING UNITS (ADU)                                                                                                                                            | PATIENT SPECIFIC UNIT-DOSE                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Description                          | Medications are dispensed in bulk into bins in medication room and are used by nurses as needed                               | Each medication is dispensed from a different locked bin. Computer unlocks the cabinet with ordered drug                                                                    | Medications are dispensed into ready-to-administer packages specific to a patient                      |
| For whom does the pharmacy dispense? | Entire unit, floor or ward                   | A specific unit or ward                                                                  | A specific patient  |
| Safety                               | <b>LOW</b><br>Nurse can grab any medications to administer to the patient. Pharmacy cannot confirm medications that were used | <b>MEDIUM</b><br>Better than traditional since automated dispensing units only allow the cabinet with correct medication to be opened, but the dose not checked by pharmacy | <b>HIGH</b><br>Pharmacy checks each patient's dispensed medication against patient profile             |

# MEDICATION DISPENSING METHODS

## Traditional



## Automated Dispensing Unit



## Patient Specific Unit-Dose



# MEDICATION DISPENSING METHODS

## UNIT DOSING



## PATIENT SPECIFIC UNIT DOSING





# MEDICATION DISPENSING TECHNOLOGY



Swisslog® PillPick®



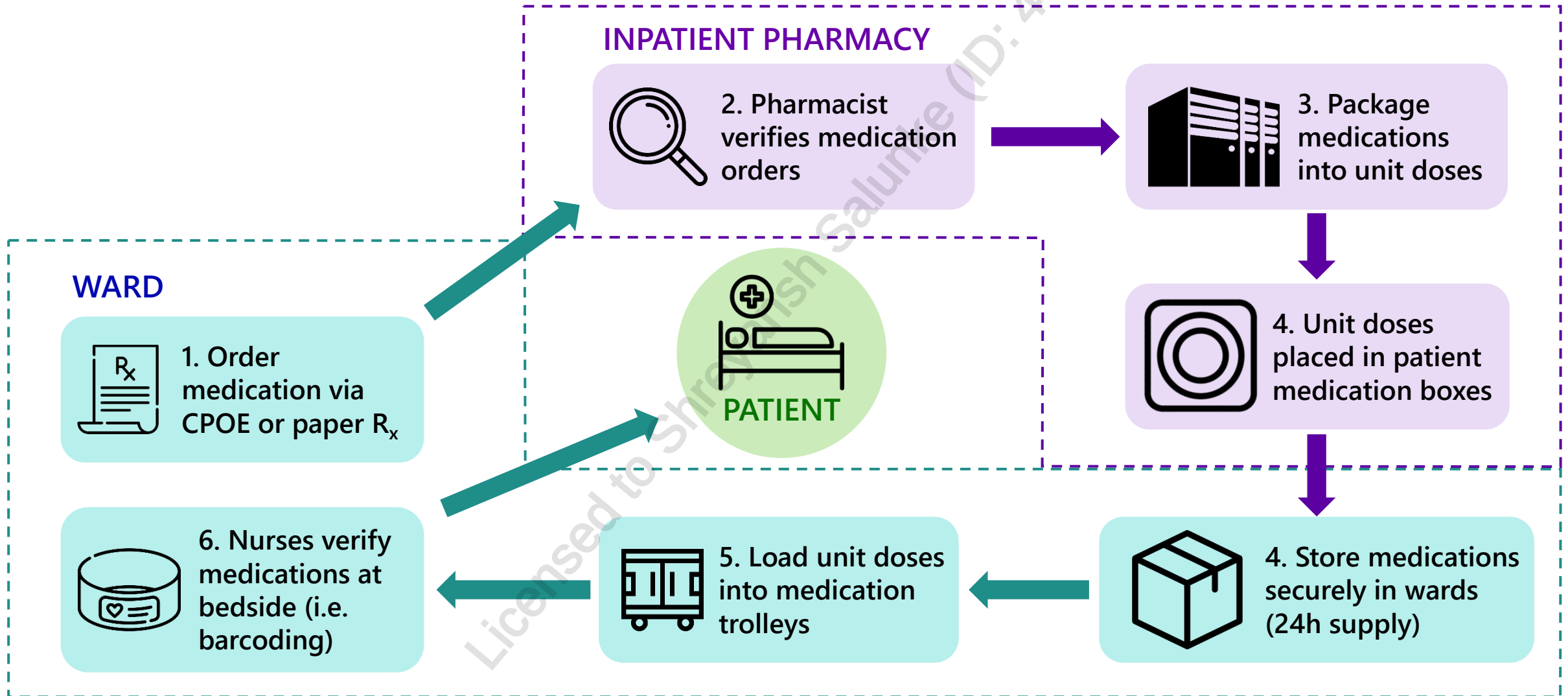
Automated Guided Vehicles

# MEDICATION ADMINISTRATION

- Administered medications are documented on a Medication Administration Record (MAR)
- CPOE orders are electronically transmitted to the electronic MAR once verified by a pharmacist
- Paper orders will be written on a paper MAR by a nurse or unit clerk once verified by a pharmacist
- Barcoding helps to ensure the correct medication is administered

[illegible]

# IDEAL DRUG DISTRIBUTION MODEL



# PATIENT CARE

Qualifying Exam Preparatory Course



**Pharm***Achieve*



# COMMUNICATION

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- Check with the care team before speaking to patients
- Wait for an appropriate time to approach patients
- Introduce yourself
- Recognize the patients may be sedated, tired, feeling unwell etc.
- Use appropriate language
- Use visuals when appropriate
- Be compassionate and empathetic

# INFECTIOUS DISEASE PRECAUTIONS

Contact Precautions

STOP

Visitor must



Wash hands

Staff must



Wash hands



Gown



Glove

Patients must (When outside the room)



Wash hands



Gown

Droplet Precautions

STOP

Visitor must (When outside the room)



Wash hands



Mask

Staff must



Wash hands



Gown



Glove



Mask with Shield

Patients must (When outside the room)



Wash hands



Mask

Airborne Precautions

STOP

Visitor must (When outside the room)



Wash hands



Mask

Staff must



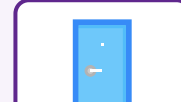
Wash hands



Negative pressure room



N95 mask



Keep door close

Patients must (When outside the room)



Wash hands



Mask

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# DONNING AND DOFFING PPE

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## Before Patient Contact

### Donning PPE

- Perform hand hygiene
- Put on the gown
- Put on mask
- Put on protective eyewear
- Put on gloves (ensure they are over the cuffs of the gown)

## After Patient Contact

### Doffing PPE

- Remove gloves
- Remove gown
- Remove eyewear
- Remove mask
- Perform hand hygiene

# INTUBATION AND VENTILATION

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## INTUBATION

- Breathing tube inserted down the patient's trachea into the lungs
- A ventilator is required to breathe for the patient
- Medication infusions for sedation, blood pressure and heart rate control are often required

## VENTILATION

- Breathing machine that breathes for the patient
- The ventilator can be hooked up to a breathing tube or a tracheostomy tube (invasive ventilation) or via a tight-fitting face mask (non-invasive ventilation)
- Sedation is not necessary unless the patient is intubated

# INTUBATION AND VENTILATION

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Intubation



Ventilation



# TAKEAWAY

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- Clinical pharmacist services include checking medication orders, completing medication reconciliation, developing pharmaceutical care plans and monitoring medications among others
- Pharmacist non-clinical services include drug safety audits, conducting drug use evaluations and maintaining resources
- Dispensing medications throughout the hospital can be done via the bulk bins, automated dispensing units, or patient specific unit dose dispensing
- Compounding may be done by nurses or pharmacy and may be sterile or non-sterile compounding
- PPE is required before entering some patient's rooms to protect against infection
- Most patients in hospital have IV access which can be used for medication administration



# CASE STUDY

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GY is a hospital pharmacy intern who is shadowing you as the hospital pharmacist. You work in a hospital in Canada, where pharmacists are integrated into multidisciplinary teams and actively participate in patient care rounds. The hospital recently transitioned from a traditional ward stock distribution system to a hybrid distribution system, incorporating automated dispensing units (ADUs) for selected wards and patient specific unit-dose distribution for most inpatient medications. High-alert medications, chemotherapy, and sterile preparations are compounded centrally in the pharmacy in accordance with the National Association of Pharmacy Regulatory Authorities (NAPRA) compounding standards. Last year, the hospital also switched from paper-based medication charts to an electronic medical record (EMR) system, which interfaces with computerized physician/provider order entry (CPOE) and allows for real-time pharmacist order verification. During your shift, some of the duties you are responsible for include verifying medication orders, checking automated dispensing cabinet refill reports, checking total parenteral nutrition prepared in the sterile compounding room by pharmacy technicians, and answering a nurse's question on barcode medication administration. Whilst performing your usual activities during your shift, GY asks you several questions about your role and hospital pharmacy distribution in Canada.

# CASE STUDY

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1. Which pharmacy practice model is being followed in this hospital?
  - a) Clinical practice-centred
  - b) Drug distribution-centred
  - c) Integrated drug distribution/clinical practice
  - d) Separate clinical and distributive practice
  
2. Which of the following statements is FALSE regarding the benefits of patient specific unit-dose dispensing?
  - a) Reduces medication errors
  - b) Allows better drug-use tracking
  - c) Reduces nurses time spent on medication preparation
  - d) Decreases pharmacy workload by eliminating repackaging requirements

# CASE STUDY

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3. Which of the following is NOT an advantage of using computerized physician order entry (CPOE)?
- a) Reduces prescriber errors with improved legibility
  - b) Provides automatic alerts for drug interactions, allergies, and renal dose adjustments
  - c) Eliminates need for pharmacist order verification
  - d) Improves workflow efficiency by eliminating paper orders
4. Which of the following is NOT considered a clinical service in hospital pharmacy?
- a) Renal dose adjustments
  - b) Best Possible Medication History (BPMH)
  - c) Adverse drug event reporting
  - d) Patient education of medications

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# CHANGE LOG

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## July 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slide 19: Added Takeaway
- Slides 20, 21: Added case study and questions
- Slide 22: Updated references

## September 2025

- Significant changes made. Addition of clinical content throughout the lecture

## March 2026

- Slide 2: Updated NAPRA competencies
- Minor formatting updates throughout